Applicability of the Oh–Kim two-spin density matrix to the Hubbard–Rashba BdG model

Scientific assessment of Eq. (19) of Oh and Kim for the model of Hutchinson, Hirsch, and Marsiglio

Reproducible analysis of the repository calculation

7 November 2026

Abstract

The Oh–Kim two-spin density matrix is applicable to the archived zero-spin-orbit-coupling calculation, but translational invariance alone is not a sufficient justification. The required assumptions are a spatially uniform lattice state, pairing at zero center-of-mass momentum, a Gaussian BdG mean-field state, a spin-diagonal normal correlator, a spin-singlet anomalous correlator, and normalization conditional on detecting one electron at each of two positions. The uniform Hutchinson–Hirsch–Marsiglio Hamiltonian and its zero-momentum BdG mean field satisfy the first three assumptions. At $V_{\mathrm{SO}} = 0$, the original spin basis also satisfies the spin-diagonal and singlet assumptions. Momentum dependence or a sign change of the extended-s-wave gap, and direction dependence on a square lattice, do not break translational invariance. The same matrix cannot be applied without generalization to physical spin qubits at nonzero Rashba coupling, where the base model contains mixed singlet–triplet pairing and spin-off-diagonal normal correlations. A separate numerical issue remains in the archived anomalous input: non-negative coherence amplitudes discard the sign of $\Delta_{\mathbf{k}}$ before Brillouin-zone integration. This input issue does not invalidate the Oh–Kim matrix algebra.

Repository state analyzed: `dc7f25fc3757077a2a764fadbd726f27da4b6d4d`.
Archived numerical case: $U/t = 75$, $\Delta_t/t = 4.5$, $V_{\mathrm{SO}} = 0$, $n = 1.875$, $N_k = 100$, $T = 0$.

1 Conclusion

The defensible conclusion is:

> The Oh–Kim reduction is used only for the spatially uniform, zero-center-of-mass-momentum BdG mean-field state at $V_{\mathrm{SO}} = 0$. In that limit the normal Green function is spin diagonal and the anomalous Green function is an even-parity spin singlet. Together with Wick factorization and conditional two-electron normalization, these assumptions yield Oh–Kim Eqs. (19)–(23), including for a momentum-dependent extended-s-wave gap. The same expression is not claimed for original-spin qubits at nonzero Rashba coupling.

Thus, the matrix can be used for the present archived $V_{\mathrm{SO}} = 0$ model, subject to the equal-time normal-correlator and anomalous-coherence corrections described below. It is not sufficient to defend it solely by saying that the Hamiltonian is translational invariant.

Addendum: equal-time normal-correlator correction

Subsequent comparison with Ketterson's Eqs. (55.3), (55.10), and (55.12) established that the archived normal line $-iu_{\mathbf{k}}^2 e^{i\mathbf{k}\cdot\mathbf{r}}$ is the $t_1 > t_2$ particle component, not the equal-time occupation

component required by Oh–Kim. The latter uses $|v_{\mathbf{k}}|^2$. Therefore, this report establishes the validity of the Oh–Kim matrix form, but its original discussion of archived numerical inputs was incomplete: both the normal time-ordering branch and the anomalous phase require correction.

The code now defaults to Ketterson's equal-time convention,

$$G_<(\mathbf{r}) = +\frac{i}{2}\sum_{\mathbf{k},s}|v_{\mathbf{k}s}|^2 e^{-i\mathbf{k}\cdot\mathbf{r}}, \qquad F^*_<(\mathbf{r}) = +\frac{i}{2}\sum_{\mathbf{k},s}\frac{\Delta^*_{\mathbf{k}s}}{2E_{\mathbf{k}s}}e^{-i\mathbf{k}\cdot\mathbf{r}}, \tag{1}$$

with $\hbar = 1$. The prefactor $1/2$ averages the two degenerate helicity contributions at $V_{\mathrm{SO}} = 0$, giving $G_<(0) = n/2$; it is applied to both G and F and therefore cancels from normalized correlators and observables. The numerical results change substantially relative to the original archived u^2 implementation. For example, at $U/t = 75$, $r = 1.50754$, $\theta = 45°$, the initial values change from $F_s = 0.77855$, $C = 0.55710$, $\chi_P = 0.88621$ bits to $F_s = 0.25130$, $C = 0$, $\chi_P = 6.50 \times 10^{-6}$ bits. The complete executed comparison is provided in `reports/equal_time_greens_comparison.md` and `reports/equal_time_vs_legacy_summary.json`. This addendum supersedes any implication below that the anomalous sign was the only archived-input issue.

2 Assumptions in the Oh–Kim derivation

Oh and Kim derive the two-electron density matrix from a two-particle Green function [1]. Their derivation requires:

1. **Gaussian BCS state.** Their Eq. (7) factorizes the two-particle Green function by Wick's theorem. This is a BCS/BdG mean-field property, not an identity for an arbitrary exact interacting Hubbard state.
2. **Spin-diagonal normal propagation.** Their Eqs. (8a)–(8b) use $G_{ss'}(\mathbf{r}, \mathbf{r}') \propto \delta_{ss'} G(\mathbf{r} - \mathbf{r}')$.
3. **Singlet anomalous propagation.** Their Eqs. (9a)–(9c) use the antisymmetric spin structure $I_{ss'} = i\sigma_y$.
4. **Translational invariance.** Their discussion before Eqs. (10)–(13) uses $G(\mathbf{r}_1, \mathbf{r}'_1) = G(\mathbf{r}_1 - \mathbf{r}'_1)$ and $F(\mathbf{r}_1, \mathbf{r}_2) = F(\mathbf{r}_1 - \mathbf{r}_2)$.
5. **Diagonal spatial detection.** Their Eq. (19) sets $\mathbf{r}_1 = \mathbf{r}'_1$ and $\mathbf{r}_2 = \mathbf{r}'_2$, describing one electron detected at each of two locations.
6. **Conditional normalization.** Their Eqs. (20)–(21) normalize the resulting two-electron detection sector to a two-spin state.

3 Proof of translational invariance for the base model

3.1 Uniform lattice Hamiltonian

Let $T_{\mathbf{a}}$ translate every lattice site by a lattice vector $\mathbf{a}$:

$$T_{\mathbf{a}} c_{i\sigma} T_{\mathbf{a}}^{-1} = c_{i+\mathbf{a},\sigma}. \tag{2}$$

The normal hopping in Hutchinson *et al.* [2] is a sum over identical nearest-neighbor bonds, the Hubbard interaction has site-independent U, and the correlated-hopping coefficient Δt is site independent. Their Rashba term has the same oriented bond structure at every site. Therefore, for an infinite lattice or periodic boundary conditions,

$$T_{\mathbf{a}} H T_{\mathbf{a}}^{-1} = H. \tag{3}$$

Rashba coupling can break inversion or spin-rotation symmetry without breaking discrete lattice translations.

3.2 Zero-momentum pairing

The retained mean-field pairs have the form $c^\dagger_{\mathbf{k}\alpha}c^\dagger_{-\mathbf{k}\beta}$, with zero total momentum. Using

$$c_{j\sigma} = \frac{1}{\sqrt{N}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_j}c_{\mathbf{k}\sigma}, \tag{4}$$

the normal equal-time correlator becomes

$$\begin{aligned}\langle c^\dagger_{i\sigma}c_{j\sigma}\rangle &= \frac{1}{N}\sum_{\mathbf{k},\mathbf{k}'} e^{-i\mathbf{k}\cdot\mathbf{R}_i+i\mathbf{k}'\cdot\mathbf{R}_j}\langle c^\dagger_{\mathbf{k}\sigma}c_{\mathbf{k}'\sigma}\rangle \\ &= \frac{1}{N}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{R}_j-\mathbf{R}_i)}n_{\mathbf{k}\sigma} = G_\sigma(\mathbf{R}_j-\mathbf{R}_i).\end{aligned} \tag{5}$$

Similarly,

$$\langle c_{i\uparrow}c_{j\downarrow}\rangle = \frac{1}{N}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{R}_i-\mathbf{R}_j)}\langle c_{\mathbf{k}\uparrow}c_{-\mathbf{k}\downarrow}\rangle = F_{\uparrow\downarrow}(\mathbf{R}_i-\mathbf{R}_j). \tag{6}$$

A simultaneous translation of $\mathbf{R}_i$ and $\mathbf{R}_j$ leaves both expressions unchanged.

3.3 Momentum-dependent and sign-changing gaps are allowed

Nothing in this proof requires $\Delta_{\mathbf{k}}$ to be constant or positive. The gap

$$\Delta_{\mathbf{k}s} = \Delta_0 - \frac{\Delta_s}{4t}\epsilon_{\mathbf{k}s} \tag{7}$$

depends on relative crystal momentum; it is not a position-dependent potential or a finite-momentum pair field. Its sign changes modify $F(\mathbf{r})$ but do not invalidate $F(\mathbf{R}_1,\mathbf{R}_2) = F(\mathbf{R}_1-\mathbf{R}_2)$.

A square lattice need not be continuously rotationally invariant. Thus G and F may depend on the direction of $\mathbf{r}$. Oh–Kim Eq. (19) can be read with vector separation; continuous rotational symmetry is not required for the spin algebra.

Open boundaries, impurities, position-dependent mean fields, finite-momentum pairing, or spontaneous translational symmetry breaking would require two-position correlators. A finite Brillouin-zone quadrature grid is not, by itself, an open-boundary real-space model.

4 From Wick factorization to the Oh–Kim matrix

For a quadratic BdG Hamiltonian, the equal-time contraction has the generic structure

$$\begin{aligned}\langle c^\dagger_{2'}c^\dagger_{1'}c_1c_2\rangle &= \langle c^\dagger_{2'}c_2\rangle\langle c^\dagger_{1'}c_1\rangle - \langle c^\dagger_{2'}c_1\rangle\langle c^\dagger_{1'}c_2\rangle \\ &\quad + \langle c^\dagger_{2'}c^\dagger_{1'}\rangle\langle c_1c_2\rangle.\end{aligned} \tag{8}$$

At two detected positions, translational invariance reduces the normal factors to $G(0)$, $G(\mathbf{r})$, and $G(-\mathbf{r})$, while the anomalous factor contains $F(\mathbf{r})F^*(\mathbf{r})$. With the spin structures above, this is Oh–Kim Eq. (19).

Define

$$g(\mathbf{r}) = \frac{G(\mathbf{r})}{G(0)}, \qquad f(\mathbf{r}) = \frac{F(\mathbf{r})}{G(0)}, \tag{9}$$

and write $g^2 = |g(\mathbf{r})|^2$, $f^2 = |f(\mathbf{r})|^2$. Conditional normalization gives

$$\rho_X(\mathbf{r}) = \frac{1}{\mathcal{N}}\begin{pmatrix} 1-g^2 & 0 & 0 & 0 \\ 0 & 1+f^2 & -(g^2+f^2) & 0 \\ 0 & -(g^2+f^2) & 1+f^2 & 0 \\ 0 & 0 & 0 & 1-g^2 \end{pmatrix}, \qquad \mathcal{N} = 4-2g^2+2f^2. \tag{10}$$

The microscopic momentum dependence enters through g and f; it does not change this spin algebra.

5 Algebraic proof of Werner form and positivity

Define

$$p = \frac{f^2 + g^2}{2 + f^2 - g^2}, \qquad |\Psi^-\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{11}$$

Direct expansion proves

$$\rho_X = (1-p)\frac{\mathbb{I}_4}{4} + p|\Psi^-\rangle\langle\Psi^-|. \tag{12}$$

The eigenvalues are

$$\lambda_s = \frac{1+3p}{4}, \qquad \lambda_t = \frac{1-p}{4} \quad \text{(threefold)}, \tag{13}$$

or, equivalently,

$$\lambda_t = \frac{1-g^2}{\mathcal{N}}, \qquad \lambda_s = \frac{1+g^2+2f^2}{\mathcal{N}}. \tag{14}$$

Because $G(\mathbf{r})$ is a normalized weighted sum of phases,

$$|g(\mathbf{r})| = \left|\frac{\sum_{\mathbf{k}} w_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}}{\sum_{\mathbf{k}} w_{\mathbf{k}}}\right| \le 1, \qquad w_{\mathbf{k}} \ge 0. \tag{15}$$

Hence $g^2 \le 1$, $\mathcal{N} \ge 2 + 2f^2 > 0$, $\lambda_t \ge 0$, and $0 \le p \le 1$. The conditional matrix is Hermitian, unit trace, and positive semidefinite. At $\mathbf{r} = 0$, $g = 1$, so $p = 1$ and $\rho_X(0) = |\Psi^-\rangle\langle\Psi^-|$ independently of $f(0)$.

These results prove consistency of the matrix form; they do not independently prove that a particular numerical implementation has calculated G and F correctly.

6 Application to the archived model

Requirement	Assessment	Evidence
Uniform square lattice	Satisfied for the stated infinite/periodic bulk model	Base Hamiltonian; `kspace_setup.py`
Zero total pair momentum	Satisfied	Hutchinson *et al.* Eqs. (14), (19)–(22)
Gaussian Wick factorization	Satisfied within BdG mean field, not an exact non-Gaussian Hubbard treatment	Mean-field decoupling in base paper
Spin-diagonal normal correlator	Satisfied at $V_{\mathrm{SO}} = 0$	`config.py`; degenerate helicity branches
Singlet anomalous spin structure	Satisfied at $V_{\mathrm{SO}} = 0$ for even extended-s-wave pairing	Original spin basis restored
Relative-coordinate correlators	Satisfied by the Fourier construction	`greens_functions.py`; translation proof
Conditional two-location state	Matches the intended construction	`density_matrix.py` implements Eqs. (20)–(21)
Original-spin use at nonzero Rashba coupling	**Not valid without generalization**	Sec. 7.3
Exact signed anomalous input	**Not satisfied by archived unsigned amplitudes when the gap changes sign**	Sec. 8

Table 1: Assumption-by-assessment summary. Source-code checks and analytical derivations are distinguished from the numerical tests below.

Hutchinson *et al.* explicitly state that their Rashba gap is mixed singlet–triplet; see their Eq. (31) and surrounding discussion [2].

7 Numerical checks and figures

7.1 Spatial correlators and lattice symmetry

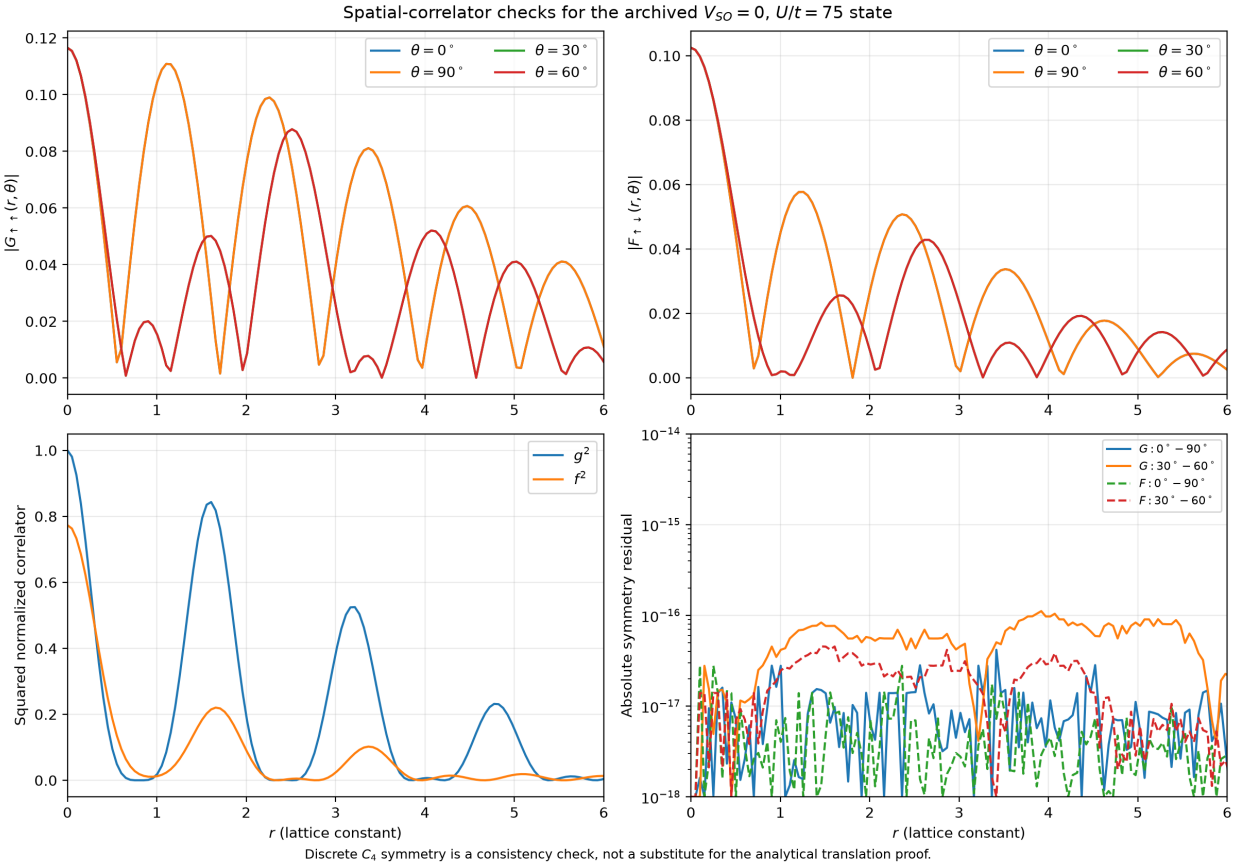


Figure 1: Archived $V_{\mathrm{SO}} = 0$, $U/t = 75$ normal and anomalous correlators. Directions related by square-lattice symmetry overlap. Maximum residuals are 4.1695×10^{-17} for $G(0°) - G(90°)$, 1.1124×10^{-16} for $G(30°) - G(60°)$, 2.7845×10^{-17} for $F(0°) - F(90°)$, and 4.5431×10^{-17} for $F(30°) - F(60°)$. These are numerical consistency checks, not replacements for the analytical translation proof.

7.2 Exact Werner equivalence and positivity

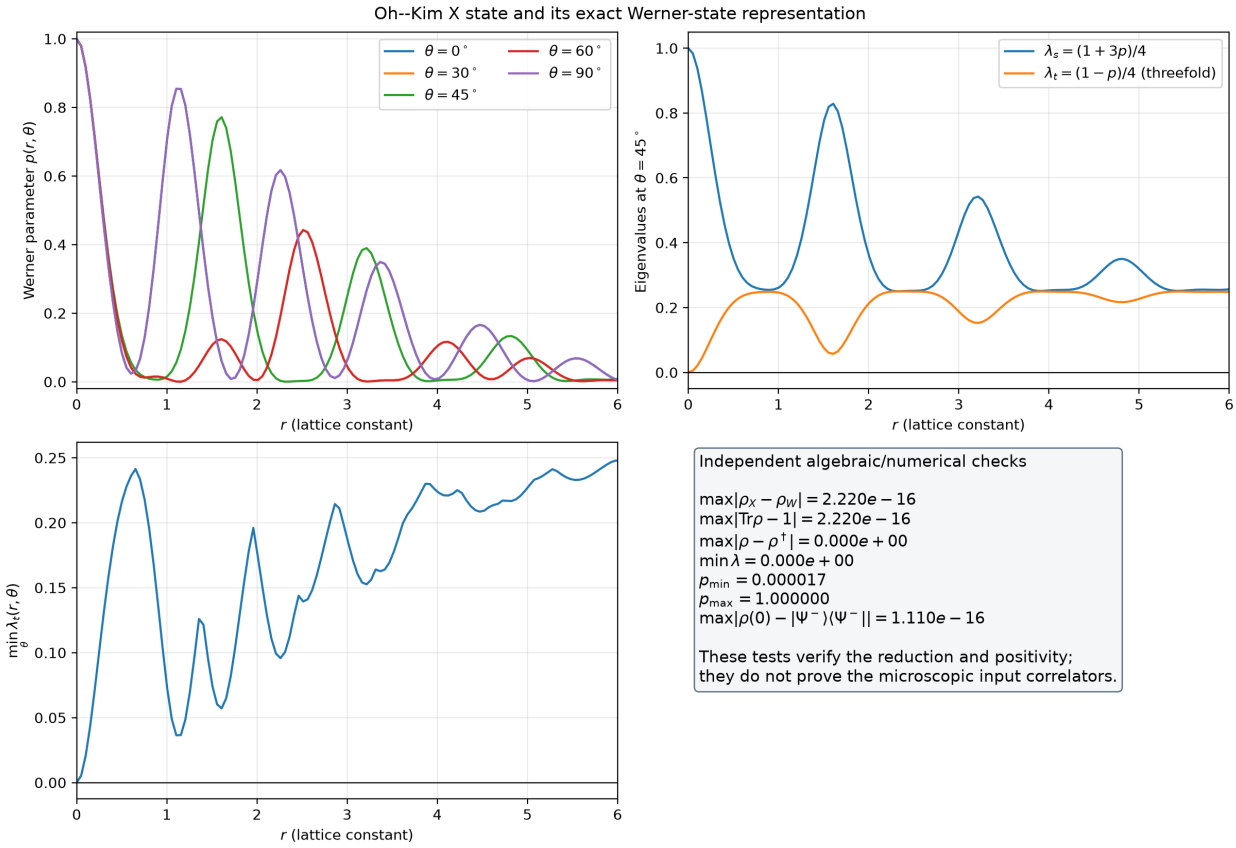


Figure 2: Werner parameter and eigenvalues from the archived correlators. Across seven angles and 200 separations, $\max|\rho_X - \rho_W| = 2.2204 \times 10^{-16}$, $\max|\operatorname{Tr}\rho - 1| = 2.2204 \times 10^{-16}$, and $\max|\rho - \rho^\dagger| = 0$. The minimum eigenvalue is zero at the pure-singlet point, not negative. The range is $1.6806 \times 10^{-5} \le p \le 1$, and $\max|\rho(0) - |\Psi^-\rangle\langle\Psi^-|| = 1.1102 \times 10^{-16}$.

7.3 Why the full Rashba case needs a different derivation

Hutchinson *et al.* define

$$c^\dagger_{\mathbf{k}s} = \frac{c^\dagger_{\mathbf{k}\uparrow} + s e^{i\theta_{\mathbf{k}}} c^\dagger_{\mathbf{k}\downarrow}}{\sqrt{2}}. \tag{16}$$

Inverting this relation gives

$$c_{\mathbf{k}\uparrow} = \frac{c_{\mathbf{k}+} + c_{\mathbf{k}-}}{\sqrt{2}}, \qquad c_{\mathbf{k}\downarrow} = e^{i\theta_{\mathbf{k}}} \frac{c_{\mathbf{k}+} - c_{\mathbf{k}-}}{\sqrt{2}}. \tag{17}$$

For $n_{\mathbf{k}s} = \langle c^\dagger_{\mathbf{k}s} c_{\mathbf{k}s} \rangle$, the original-spin off-diagonal coefficient is

$$\langle c^\dagger_{\mathbf{k}\uparrow} c_{\mathbf{k}\downarrow} \rangle = \frac{e^{i\theta_{\mathbf{k}}}}{2} (n_{\mathbf{k}+} - n_{\mathbf{k}-}). \tag{18}$$

At $V_{\mathrm{SO}} = 0$, degeneracy makes this quantity vanish. At nonzero Rashba coupling it need not vanish.

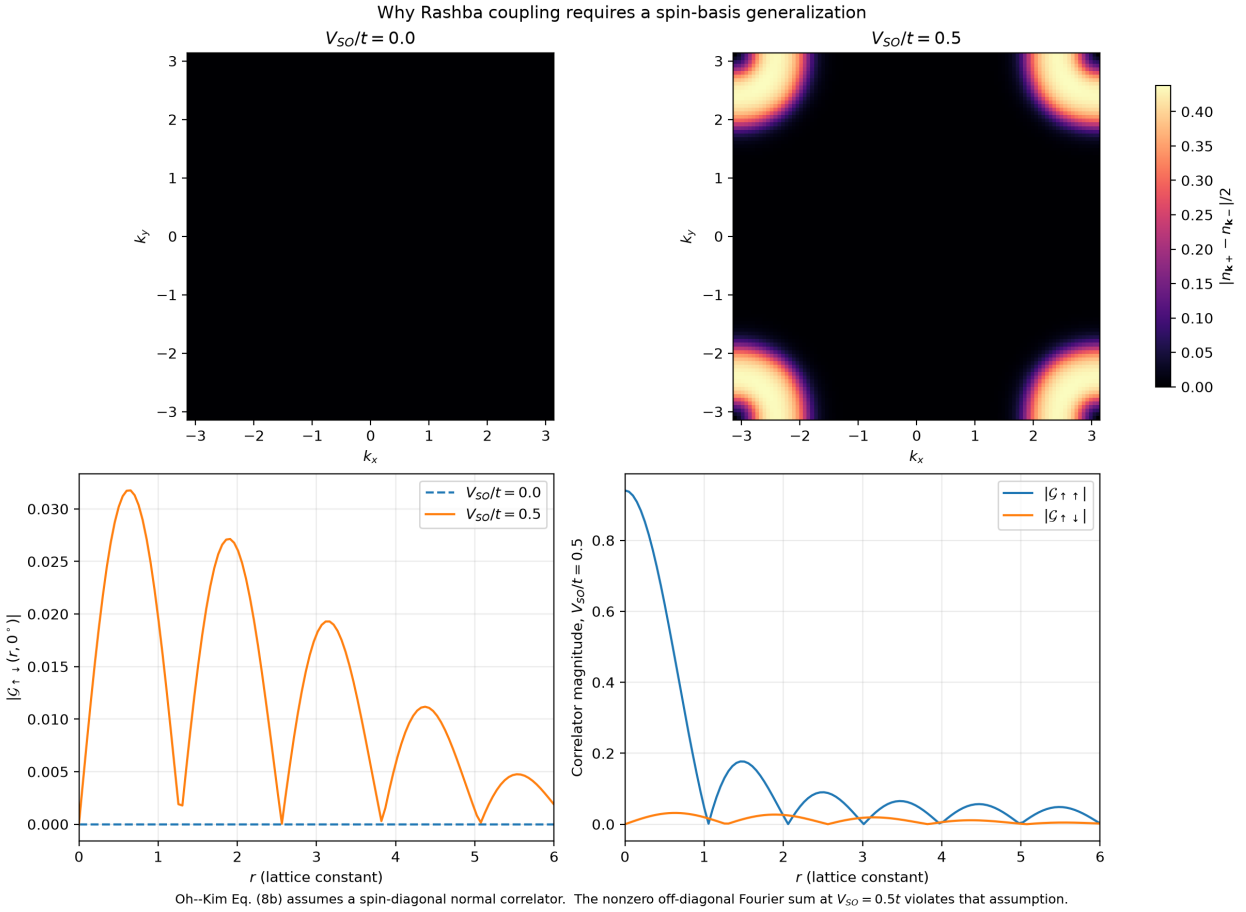


Figure 3: Original-spin normal-correlator diagnostic. The independently executed $V_{\mathrm{SO}} = 0.5t$, $U/t = 75$ control solution converged with maximum residual 3.526×10^{-12}. It gives $\max_r |\mathcal{G}_{\uparrow\downarrow}(r, 0^\circ)| = 0.0317533$, whereas the result is exactly zero at $V_{\mathrm{SO}} = 0$. This violates Oh–Kim's spin-diagonal assumption in their Eq. (8b). Translational invariance survives Rashba coupling, but the simple spin reduction does not automatically survive it.

A nonzero-V_{SO} treatment must either construct the full spin-resolved 4×4 Nambu correlator and derive a generalized two-spin state, or explicitly define helicity pseudospin qubits and justify their different operational meaning.

8 Separate issue: signed anomalous coherence

For the BdG convention stated in the manuscript,

$$u_{\mathbf{k}s} v^*_{\mathbf{k}s} = -\frac{\Delta^*_{\mathbf{k}s}}{2E_{\mathbf{k}s}}. \tag{19}$$

The archived solver constructs $u_{\mathbf{k}s}$ and $v_{\mathbf{k}s}$ as non-negative square roots, and the Green-function routine multiplies those non-negative amplitudes. When $\Delta_{\mathbf{k}s}$ changes sign, this is not generally equal to Eq. (19) before Brillouin-zone integration.

A separate comparison used the same archived normal G, momentum grid, and quadrature, replacing only the anomalous coherence by Eq. (19).

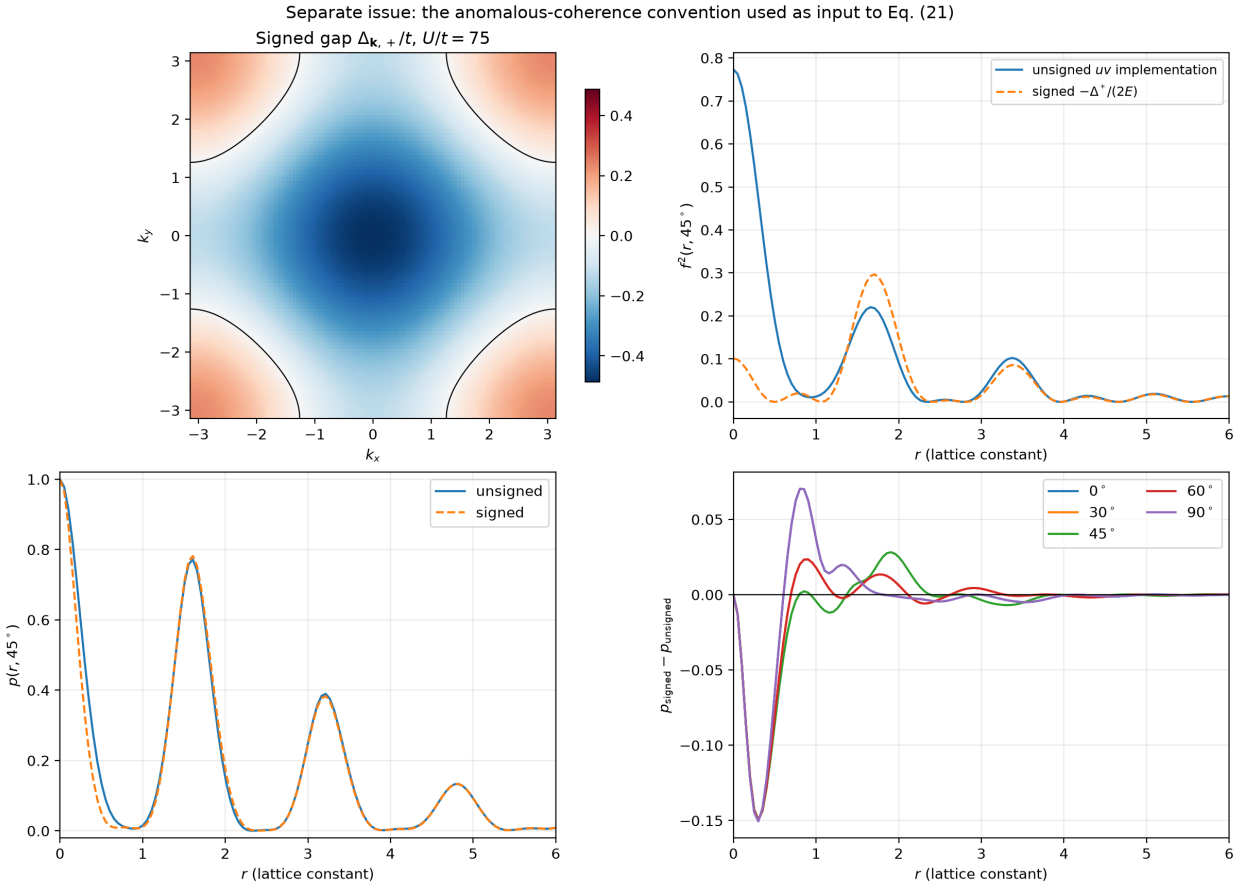


Figure 4: Effect of retaining the gap sign at $U/t = 75$, $V_{\text{SO}} = 0$. The gap changes sign in the top-left panel. Over the full angular/radial grid, $\max|F_{\text{signed}} - F_{\text{archived}}| = 0.1395489$ and $\max|p_{\text{signed}} - p_{\text{unsigned}}| = 0.1504256$. The signed-input matrices remain positive and normalized. These are new numerical results and have not been silently substituted into the archived manuscript arrays.

This discrepancy does not invalidate Oh–Kim Eqs. (19)–(23). It means that the archived numerical values cannot yet be defended as exact signed-BdG results until the anomalous input is corrected or the unsigned approximation is independently justified. The appropriate next step is to regenerate the observables using Eq. (19), not to change the density-matrix formula.

9 Scope of the proof

Established analytically

1. Uniform zero-momentum pairing yields relative-coordinate G and F.
2. Momentum dependence and gap sign changes do not break translational invariance.
3. Gaussian factorization plus spin-diagonal normal and singlet anomalous correlators yields Oh–Kim Eq. (19).
4. The normalized matrix is exactly a Werner state and is positive for $|g| \leq 1$.
5. At $r = 0$, it is exactly the singlet.

Established numerically

1. The archived arrays respect square-lattice symmetry to floating-point precision.

2. The implemented X matrix and independently constructed Werner matrix agree to machine precision.
3. All tested matrices are Hermitian, normalized, and positive semidefinite.
4. A nonzero-Rashba control calculation produces original-spin off-diagonal normal correlations.
5. Retaining the signed gap changes the anomalous input appreciably for the tested $U/t = 75$ state.

Not established

1. Applicability of the unchanged matrix to original-spin qubits at $V_{\mathrm{SO}} \neq 0$.
2. Exactness beyond Gaussian BdG mean-field theory.
3. Applicability to open boundaries, impurities, position-dependent pairing, or finite-momentum pairing.
4. Correctness of the archived unsigned anomalous correlator as an exact signed-BdG result.
5. A canonical fixed-electron-number reduction beyond the conditional two-detection normalization used by Oh and Kim.

10 Manuscript-ready justification

> In the zero-spin-orbit-coupling limit considered in the numerical calculations, the two helicity branches are degenerate and the original spin basis is restored. The spatially uniform BdG mean-field Hamiltonian contains only zero-center-of-mass-momentum pairs, so the equal-time normal and anomalous correlators depend on the relative coordinate $\mathbf{r} = \mathbf{R}_1 - \mathbf{R}_2$. Because the BdG state is Gaussian, its two-particle contraction factorizes by Wick's theorem. The normal correlator is spin diagonal and the even-parity anomalous correlator has singlet spin structure. Conditioning on one electron detected at each position and normalizing the resulting two-spin sector therefore yields the Oh–Kim matrix, Eqs. (19)–(23) of Ref. [1], including for the momentum-dependent extended-s-wave gap. This reduction is not used for original-spin qubits at nonzero Rashba coupling, where spin-off-diagonal normal correlations and mixed singlet–triplet pairing require a generalized spin-resolved construction. The anomalous spatial input is evaluated using the signed BdG coherence factor $u_{\mathbf{k}} v_{\mathbf{k}}^* = -\Delta_{\mathbf{k}}^* / (2E_{\mathbf{k}})$.

The final sentence should be included only after the numerical pipeline actually uses the signed factor.

11 Reproducibility

Run from the repository root:

```
.venv/bin/python
analysis/validate_density_matrix_scope.py
```

The script generates `reports/density_matrix_validity_checks.json`, `data/density_matrix_validity_checks.npz`, and the four figures in `figs/`. No IBM hardware or external numerical results were used. The nonzero-Rashba example is a newly executed control calculation, not part of the archived manuscript sweep.

References

[1] S. Oh and J. Kim, *Entanglement of electron spins in superconductors*, Phys. Rev. B **71**, 144523 (2005), https://arxiv.org/pdf/quant-ph/0406013.

[2] J. Hutchinson, J. E. Hirsch, and F. Marsiglio, *Enhancement of superconducting T_c due to the spin-orbit interaction*, https://arxiv.org/pdf/1801.08988.